

CRUCIBLE: A Multi-Agent Framework for Automated Annotation Rubric Calibration via Iterative Inter-Judge Disagreement Resolution

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October 2026

Abstract

Annotation quality is a limiting factor in supervised learning pipelines that rely on the LLM-as-judge paradigm. This paper’s central finding is that an automated convergence criterion can certify agreement that chance-corrected measurement does not support, and that the certification is an artifact of the agreement metric being optimized. Our weighted agreement statistic has a floor of 0.667 at zero unanimity, so a convergence threshold of $\tau = 0.80$ is satisfied at 8 of 20 unanimous items: the loop halts and declares success while 60% of items still carry judge disagreement. The domain that crossed τ on this metric at 81.7% has a chance-corrected inter-judge agreement of only $AC1 = 0.286$. We reach this finding while building and evaluating **CRUCIBLE**, a multi-agent system that automatically calibrates annotation rubrics through iterative disagreement diagnosis and targeted revision. Manual rubric refinement does not scale and introduces annotator-specific bias that usually goes unaudited and is hard to reproduce. Three heterogeneous local judge models independently evaluate a transcript corpus; a coordinator model (Claude Sonnet) analyzes per-model criterion-level disagreement patterns and proposes rubric revisions; the loop repeats until all domains converge or a fixed iteration budget is exhausted, flagging any domain whose agreement stagnates. CRUCIBLE was developed to solve a concrete annotation bottleneck in building a Signal-to-Noise Ratio (SNR) classifier for YouTube educational content, motivated by an earlier, differently-scoped pilot (a frontier chat-model ensemble applying a domain-agnostic rubric, reported in the companion SNR paper) that showed substantial, inconsistent inter-judge disagreement. Applied to 60 YouTube transcripts across three content domains—General Education, Technology & AI, and Career & Self-Improvement—CRUCIBLE reached 96.7%, 81.7% and 70.0% weighted inter-judge agreement, corresponding to chance-corrected AC1 of 0.929, 0.286 and -0.080 . An independent human validation study covering the full corpus—20 transcripts per domain, scored by two disjoint cohorts of three annotators each (18 annotators)—finds that judge and human chance-corrected reliability rank the three domains identically, so the judges find difficult what humans find difficult. We report as a negative result that agreement between human consensus and the system’s labels carries almost no evidential weight in this design: five of six domain-round cells fail to beat a majority-class baseline that labels every item with the domain’s dominant class, and the pooled rate of 75% is below the corresponding baseline. Against that baseline, a between-domain contrast we previously reported as preliminary evidence of a human-alignment dissociation does not survive, and we retract it. Because raw agreement is inflated by label prevalence, we report chance-corrected agreement (Fleiss’ κ and Gwet’s AC1) throughout; these locate CRUCIBLE’s value not in raising raw agreement, and not in match against human consensus, but in diagnosing whether residual disagreement is rubric-fixable, capability-bounded, or

prevalence-dominated—and in the accompanying warning that a stopping rule defined on an uncorrected agreement metric will terminate before agreement has been achieved. We additionally check whether the calibrated rubrics’ effect on agreement generalizes to a 90-transcript corpus never seen by the coordinator during revision; the weighted-agreement metric improves there too, though the underlying majority-vote annotations themselves do not change on any of the 90 transcripts (Section 5.3; see Limitations for the precise sense in which this corpus is unseen). A pre-run discriminative-validity screening check identified a degenerate judge (Gemma 2 9B, producing identical LOW labels for all 60 transcripts) during development, before the calibration run reported here; it was replaced with Qwen 2.5 7B. The calibrated rubrics, applied by a separate frontier ensemble (reported in the companion paper), produced labels that trained a binary classifier on the SNR task achieving $F1 = 0.871$ with $\text{recall} = 0.964$ on the HIGH-signal class as a downstream case study. Code, data and the script reproducing the human validation analysis are available at <https://github.com/biditdas18/rubric-calibration-agent>.

Note on this version. This version supersedes the earlier preprint. The human validation study has been extended from 10 to 20 transcripts per domain (the full corpus, scored by two disjoint annotator cohorts), and the analysis now reports a majority-class baseline beside every human–system match rate. Two changes follow from that. First, the between-domain match contrast the earlier version offered as preliminary evidence of a human-alignment dissociation does not replicate and is retracted; judge and human chance-corrected reliability instead rank the three domains identically. Second, the paper’s central claim has moved to a property of the convergence criterion itself (Section 5.1.1), which was derivable from the original data and which we had not drawn out. Analyses that score against the original 30-item human reference—the frontier-ensemble comparison, the rubric-portability test, the reasoning-before-label ablation, the rejection-sensitivity check and the channel-clustered intervals—were not re-run at the larger sample and are marked where they appear.

1 Introduction

The LLM-as-judge paradigm has become a standard component of annotation pipelines in settings where human labeling is expensive, slow, or subject to individual bias [1]. A single LLM prompted with an evaluation rubric can produce labels at scale, but single-model annotation carries systematic biases rooted in the model’s training distribution, instruction-following style, and boundary-case interpretation [2]. Multi-model ensembles reduce this risk, but introduce a new dependency: the rubric must be precise enough that multiple independent models reach consistent conclusions. A poorly specified rubric produces disagreement not because the content is genuinely ambiguous, but because different models resolve underspecified criteria differently.

This paper originates from a concrete failure of that kind. We were building a Signal-to-Noise Ratio (SNR) framework¹ for binary classification of YouTube educational video transcripts—separating substantive instructional content (HIGH signal: concrete actionable steps, methodological coherence, named frameworks) from low-value filler (LOW signal: generic motivational advice, fear-based framing, promotional material, repetition without progression). The task required labeled training data reliable enough to train a lightweight classifier deployable at inference scale.

During development, an earlier annotation pipeline—a differently-scoped, differently-modeled experiment reported in the companion SNR paper [24]—showed substantial and inconsistent dis-

¹The SNR classifier is described in a companion paper: Das, B. (2026). Signal-to-Noise Ratio Detection in Educational Video Content: A Binary Classification Framework Using Transcript Embeddings. SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6866745. That work cites CRUCIBLE as its annotation methodology.

agreement across judges, motivating the need for systematic rubric diagnostics.² Two remedies were available: (1) manually label a gold dataset per domain, or (2) automatically calibrate the rubrics until the ensemble converges. Option 1 was rejected on two grounds. First, reviewing transcripts of 10–15 minute videos at the scale required is prohibitively time-intensive. Second, and more fundamentally, manual annotation introduces annotator-specific bias that is reducible only through multiple annotators and adjudication: an annotator who values career development content intrinsically applies a systematically different threshold than one who does not. Without a multi-annotator comparison, that bias goes unaudited and undisclosed—baked into every label silently.

We designed CRUCIBLE to solve this. It is a coordinator-mediated, iterative calibration loop requiring no human reference labels and no domain expertise beyond an initial rubric draft. Three heterogeneous local models serve as judges; a frontier coordinator model diagnoses their disagreement patterns and proposes targeted rubric revisions; the loop repeats until convergence or a fixed iteration budget is exhausted. The resulting domain-specific rubrics are versioned and auditable. The labels they produce are silver quality—not gold—but empirically validated against a measurable inter-judge criterion rather than a single annotator’s unchecked judgment.

The downstream classifier serves as a case study in whether the calibrated labels are usable at scale, not as an independent validation of their agreement with the available human reference (Section 5 and Limitations address that question directly). The rubrics produced by CRUCIBLE were applied by a separate frontier ensemble (GPT-4o mini, Gemini 2.5 Flash, Llama 3.3 70B; reported in the companion paper) to label transcripts; an SVM trained on 389 transcripts (89 real, labeled via this calibrated-rubric ensemble methodology, plus 300 synthetically generated transcripts with labels assigned by construction, intended to match the generating prompt’s target class) achieved $F1 = 0.871$ with $\text{recall} = 0.964$ on the HIGH-signal class, evaluated on a disjoint held-out set of 60 real transcripts, compared to a majority-class baseline of $F1 = 0.000$ and, in a synthetic-only training ablation reported in the companion paper [24], $F1 = 0.747$ when trained on the 300 synthetic transcripts alone—indicating the real, calibrated-rubric-labeled transcripts contribute materially beyond what synthetic data alone provides. We did not attempt training on labels from the uncalibrated pilot pipeline described above; the majority-class baseline ($F1 = 0.000$) indicates the task itself is non-trivial.

Beyond the calibration loop and this downstream case study, a set of follow-up analyses—some conducted after the initial calibration run, one (judge replacement, Section 6.3) run as a direct, controlled test of this paper’s own prescriptive claim—surfaced this paper’s central finding: CRUCIBLE’s own convergence signal is not by itself sufficient evidence of label quality. This paper is organized around two questions those analyses answer: **(1)** does disagreement-driven rubric revision generalize past the transcripts it was tuned on, and **(2)** does higher inter-judge agreement correspond to higher agreement with human judgment. An external human validation study, a frontier-model rubric-transfer test, a reasoning-before-label ablation, and a direct judge-replacement test each speak to question (2); an evaluation on a 90-transcript corpus unseen by the coordinator during revision speaks to question (1). We report these alongside the calibration results because they qualify how those results should be read, and we return to both questions explicitly in Sections 5–6.

²That pilot used a domain-agnostic rubric applied across a frontier chat-model ensemble (GPT-4, Gemini, Meta AI), distinct from the local judge panel (Llama 3.1 8B, Mistral 7B, Qwen 2.5 7B) used for calibration in this paper, and a 3-category (HIGH/MID/LOW) label space rather than the binary space CRUCIBLE’s own weighted- A metric (Eq. 1) operates in. See Appendix B for the exact figures and why they are not directly comparable in magnitude to CRUCIBLE’s own convergence numbers.

The core contributions of this paper are:

1. **A convergence criterion can certify agreement that chance-corrected measurement does not support.** The weighted agreement statistic the loop optimizes has a floor of 0.667, so the threshold $\tau = 0.80$ is met once 40% of items are unanimous, leaving 60% carrying judge disagreement (Section 5.1.1). The domain the loop certified as converged at 81.7% has a chance-corrected inter-judge agreement of 0.286. The general form—required unanimity $(\tau - f)/(1 - f)$ for a statistic with floor f —applies to any thresholded agreement measure whose scale is not established, including raw percent agreement.
2. **Raising the optimized metric does not raise agreement with the human reference.** Three interventions move the metric without moving human agreement: a frontier-model baseline (Section 5.8) shows the calibrated rubric does not transfer to a different judge model, instead teaching it to mimic the local ensemble’s specific errors; a reasoning-before-label ablation (Section 5.8.3) shows explicit reasoning raises inter-judge agreement without raising agreement with the human reference; and a judge-replacement test (Section 6.3) raises inter-judge agreement from 0.700 to 0.883 while leaving the panel’s majority-vote output completely unchanged—though that last result is mechanically guaranteed, since the two retained judges already agreed on 90% of this domain’s transcripts. All three score against the original 30-item human reference and are related diagnostic analyses on one reference set, not independent replications.
3. **A human validation study covering the full corpus, reported as a negative result.** Twenty transcripts per domain scored by two disjoint cohorts of three annotators (Section 5.6) show judge and human chance-corrected reliability ranking the three domains identically—the judges find difficult what humans find difficult. Match against human consensus, by contrast, fails to beat a majority-class baseline in five of six domain-round cells, so it carries almost no evidential weight in this design. A between-domain match contrast reported in the earlier version of this work as preliminary evidence of a human-alignment dissociation does not replicate, and we retract it.
4. **An automated rubric calibration loop** with coordinator-mediated disagreement diagnosis and targeted revision. Unlike prior rubric-generation and refinement systems, which are validated against reference judgments or accuracy signals, CRUCIBLE studies a narrower setting: iterative rubric revision driven solely by inter-judge disagreement, with no human reference labels used during calibration.
5. **A held-out generalization check.** Re-scoring both the initial and final calibrated rubrics on a 90-transcript corpus never shown to the coordinator during revision (Section 5.3) shows the revision’s effect on the weighted inter-judge *agreement metric* generalizes past the calibration corpus for the two domains that were actually revised, with a positive, CI-excluding-zero effect for Career & Self-Improvement specifically—though, as detailed there, this generalization is driven disproportionately by one judge’s behavior shifting, not uniform three-way convergence, and the underlying majority-vote annotations do not change on any of the 90 transcripts under either rubric.
6. **The capability-ceiling finding, revised.** The system empirically distinguishes rubric-fixable disagreement (Technology & AI: 76.7% $\rightarrow$ 81.7% in one iteration) from disagreement that further rubric revision did not resolve (Career & Self-Improvement: failed to converge despite two revisions, with its best rubric at 70.0%). We initially attributed this to a pure model-capability limitation. The enlarged human study (Section 5.6) instead shows that Career’s human annotators agree with each other barely above chance themselves (AC1 = 0.24 pooled over 20 items,

95% CI $[-0.06, 0.56]$), the lowest of the three domains and matching the judge panel’s own ordering. The domain is therefore one where the substantive/generic distinction is weakly determined for judges and humans alike, rather than one where the judges are misaligned with a well-defined human standard. We cannot fully separate this from a structural property of the panel itself: our judge-replacement test (Section 6.3) found the two static judges already agreed with each other on 90% of this domain’s transcripts, so no third judge could have changed the panel’s majority-vote output on this set regardless of the ceiling’s underlying cause.

7. **Per-domain convergence.** Domains converge independently, each against its own rubric and disagreement history. Two supporting quality-control components are detailed in Sections 3.5–3.6: a discriminative-validity pre-screening gate that caught and removed a degenerate judge (Gemma 2 9B, zero discriminative outputs) before the reported run, and a columnar evaluation order that made repeated 60-transcript, 3-model passes tractable on consumer hardware. Each iteration also re-evaluates any domain that converged on the immediately preceding iteration, as a one-time check that its agreement rate holds on a second pass with the same rubric and judges (Section 6.5)—an output-stability check, not a cross-domain interference test, since coordinator calls are single-domain, stateless API requests with no mechanism for one domain’s revision to affect another’s.
8. **A downstream case study.** The calibrated rubrics, applied by a separate frontier ensemble, produced labels that trained an SVM classifier achieving $F1 = 0.871$ and $\text{recall} = 0.964$ on the HIGH-signal class (Section 5). We treat this as a usability case study, not an independent validation of the labels’ agreement with the available human reference: Section 5.7 shows the same frontier ensemble tracks human judgment about as loosely as the local calibration panel does.

2 Related Work

2.1 LLM-as-Judge

The use of large language models as automated evaluators was formalized by Zheng et al. [1] in the MT-Bench benchmark. Subsequent work extended this paradigm to structured rubrics [3], fine-grained multi-dimensional scoring [4], and customizable evaluation dimensions [5]. A persistent finding is that single-model judges exhibit systematic biases—position bias, verbosity bias, and self-preference—that reduce reliability [1, 2]. Multi-model ensembles have been proposed to mitigate these biases [6], but ensemble reliability still depends on rubric quality.

2.2 Rubric Design and Calibration

LLM-Rubric [7] combines LLM predictions with a small neural network to predict human judge annotations. AdaRubric [8] proposes task-adaptive rubric generation for agent evaluation. Autorubric [9] provides a unifying framework for rubric-based evaluation across criterion types. Prior rubric-refinement work that does iterate does so against human-labeled reference data [10]. Our narrower claim relative to this body of work is not that automated rubric revision itself is unaddressed in general, but that we revise domain-specific annotation rubrics using cross-model disagreement as the sole feedback signal, with no human reference labels used during calibration, while explicitly investigating when that signal produces panel-specific consensus rather than alignment with human judgment (Sections 5.6–5.8.3) — a question the reference-judgment-validated systems above are not positioned to ask, since their validation signal is human-labeled by construction. SciIF [11] pairs

problems with a fixed catalog of scientific constraints and structured verification rubrics, but treats the rubric as fixed rather than something to be revised. Recent work also automates rubric generation and refinement: RRD [17] recursively decomposes and filters rubric criteria, and GER-Eval [18] has LLMs design and apply their own rubrics. Both are still validated against reference judgments or accuracy signals; CRUCIBLE instead drives revision purely from inter-judge agreement with no human reference labels, and separately audits, via human validation, when that agreement-only signal is and is not trustworthy.

2.3 Disagreement Diagnosis

Related self-deliberation mechanisms appear in evaluation frameworks such as AraTable [12], where a judge re-examines its own assessment. Multi-agent debate [13] resolves disagreement through structured exchange among models. These approaches are self-revision or debate; neither produces a generalized rubric revision that improves agreement across the full corpus and on future evaluation runs.

2.4 Multi-Agent Coordination

Multi-agent debate frameworks [13] have been applied to improve LLM output reliability through structured disagreement resolution. CRUCIBLE applies the coordinator-mediator pattern asymmetrically: judge models do not communicate with each other; only the coordinator sees aggregate disagreement patterns and produces the revision.

2.5 Annotation Bias and Scalability

The annotation bottleneck has been addressed through weak supervision [14], programmatic labeling [15], and knowledge distillation from frontier models [16]. CRUCIBLE is complementary: it produces rubrics calibrated against inter-judge agreement, grounding ensemble annotation at scale—though, as Section 5.6 shows, that calibration signal alone does not guarantee the rubrics are human-validated.

3 The CRUCIBLE System

3.1 Problem Statement

Given a corpus $\mathcal{D} = \{d_1, \dots, d_N\}$ to be annotated with a binary label $y \in \{H, L\}$, and an initial rubric $R^{(0)}$ specifying criteria for each label, we seek a revised rubric $R^{(k)}$ such that the weighted inter-judge agreement $A(R^{(k)}, \mathcal{D}) \geq \tau$ for a target threshold τ , with the revision process requiring no human-labeled reference data.

The key insight is that inter-judge disagreement encodes rubric underspecification: when two competent models applying the same rubric reach opposite conclusions on the same document, the rubric has not resolved the case. A coordinator with visibility into both decisions and the criterion-level justifications can identify *which criterion was misapplied and how*—and propose a targeted revision to close the gap.

3.2 Architecture

Figure 1 illustrates the full system. Judge models operate independently and in isolation from each other; only the coordinator sees their combined outputs. The rubric revision loop is the only

feedback path—there is no human in the loop after initialization.

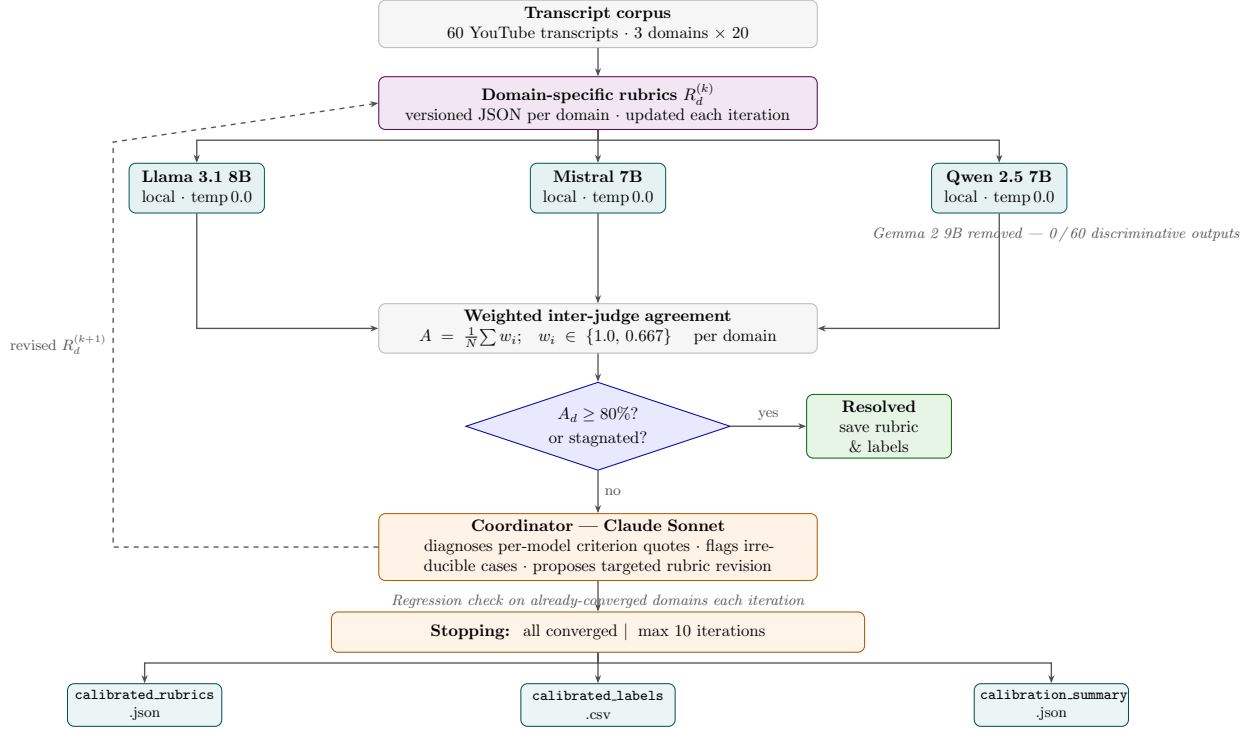


Figure 1: CRUCIBLE system architecture. Judge models operate independently at temperature 0.0. The dashed arrow is the only feedback path: revised rubrics flow back into the next iteration. No human input occurs after initialization.

3.3 System Components

Judge models. Three heterogeneous local models running at temperature 0.0 via Ollama: Llama 3.1 8B (Meta), Mistral 7B (Mistral AI), and Qwen 2.5 7B (Alibaba). Temperature 0.0 yields near-deterministic outputs, isolating rubric quality as the variable of interest. The three models were chosen to represent different training lineages, maximizing the diagnostic value of their disagreements. These local models serve as the calibration instrument only; production labeling in the downstream application uses a separate, stronger frontier ensemble (companion paper).

Coordinator model. Claude Sonnet (claude-sonnet-4-6) [22], via API. The coordinator receives: (a) the current rubric, (b) per-model label assignments and criterion-level justification quotes for each disagreement case, and (c) disagreement counts per criterion. It produces a structured JSON revision proposal specifying which criterion to modify and the revised text.

Agreement metric.

$$A = \frac{1}{N} \sum_{i=1}^N w_i, \quad w_i = \begin{cases} 1.0 & \text{all three judges agree} \\ 0.667 & \text{exactly two judges agree} \end{cases} \quad (1)$$

For three judges on a binary label, every item is either unanimous ($w = 1.0$) or a two-one majority ($w = 0.667$); the no-majority case cannot occur, so A is bounded below by 0.667. Under independent judges the expected value of A is approximately 0.75, which should be read as the chance level for this metric; label skew can raise it further.

All-agree rate (unanimous three-way agreement) is tracked separately as a stricter secondary metric.

3.4 Calibration Loop

Initialization. Domain-specific rubrics $R_d^{(0)}$ are written at moderate abstraction—sufficient to convey intent but not yet precise enough to resolve edge cases consistently.

Iteration k :

1. All three judges evaluate all documents using $\{R_d^{(k)}\}$.
2. Weighted agreement $A_d^{(k)}$ is computed per domain.
3. For each non-converged, non-stagnated domain with $A_d^{(k)} < \tau$:
 - a. Extract disagreement cases (at least one judge diverged).
 - b. Assemble per-model criterion quotes for each case.
 - c. Coordinator diagnoses and proposes revision.
 - d. If valid JSON: $R_d^{(k+1)} \leftarrow \text{revise}(R_d^{(k)})$.
 - e. Irreducible cases (genuine ambiguity, not rubric failure) are logged and excluded from stagnation counting.
4. Regression check on each already-converged domain.

Stopping. A domain is marked *stagnated* when $\Delta A_d \leq 0$ for three consecutive iterations; it is then excluded from further revision and held at its current rubric, but this does not terminate the loop. The loop terminates only when (a) all domains have converged or (b) the maximum of 10 iterations is reached. Because Career & Self-Improvement never converged, the present run continued to the 10-iteration cap, with converged domains monitored by regression checks and the stagnated domain held in place.

3.5 Judge Quality Control

Prior to the primary run, each candidate judge is evaluated for discriminative validity. A degenerate judge—one assigning a single label to all documents—provides zero inter-judge signal and pollutes agreement scores.

During development, Gemma 2 9B produced LOW labels for all 60 transcripts across all domains (0% HIGH rate). This is consistent with treating LOW criteria as a checklist to confirm rather than a comparative threshold to apply. Gemma 2 9B was replaced with Qwen 2.5 7B before the calibration run reported here.

3.6 Columnar Evaluation Order: An Engineering Optimization

The order in which judge models are invoked across the corpus has a dramatic effect on runtime: batching evaluation by model rather than by document (running each judge across all 60 transcripts before switching models) reduces model-load events from 180 to 3 per iteration and cuts per-iteration runtime from an incomplete >30 minutes to 17–20 minutes on consumer hardware, making the full 10-iteration run tractable. This is an engineering detail, not a scientific finding; the full derivation and the naive row-wise alternative it replaces are given in Appendix C for readers interested in reproducing the setup.

4 Experimental Setup

Task and corpus. The target task is binary classification of YouTube educational video transcripts under the SNR taxonomy: HIGH signal (substantive instructional content: concrete actionable steps, methodological coherence, named frameworks) vs. LOW signal (generic motivational advice, fear-based framing, promotional material, repetition without progression). The corpus comprises 60 transcripts across three domains (20 per domain): General Education, Technology & AI, and Career & Self-Improvement.

Sampling protocol. Transcripts were collected via two methods: approximately 20 via manual search, and the remaining approximately 130 (spanning both this 60-transcript evaluation corpus and the 90-transcript training-supplement corpus used in Section 5.3) via the YouTube Data API v3, using a VPN to work around geographic and rate-limit restrictions. API search was constrained to medium-length videos (4–20 minutes) with closed captions available, filtered for English relevance, using domain-specific query sets (5 base queries per domain, expanded with “tips”/“tutorial” variants when needed to fill the candidate pool); candidates were deduplicated against the existing evaluation set and filtered to 300–8000 words per transcript. Candidate selection prioritized a fixed allow-list of known channels per domain (established educational/technical channels expected to produce clear HIGH signal) before falling back to general search results—a deliberate design choice trading naturalistic diversity for more consistent signal quality, and a limitation on how representative the corpus is of YouTube educational content generally (see Limitations). This allow-list bias plausibly contributes to the strong HIGH-signal skew reported below (**Corpus label distribution**): a channel allow-list built to surface clear HIGH-signal content is, by construction, unlikely to also surface a representative share of genuinely LOW-signal content, so the corpus is better described as a curated educational-content sample than as representative of YouTube educational content generally.

Corpus label distribution. Both corpora are strongly HIGH-skewed at the majority-vote level. On the 60-transcript evaluation corpus (final calibrated labels), majority-vote HIGH rate is 100% for General Education, 60% for Technology & AI, and 90% for Career & Self-Improvement; the individual judges are more HIGH-skewed still (Llama 3.1 8B: 100%, 85%, 100% HIGH by domain; Mistral 7B: 100%, 60%, 90%; Qwen 2.5 7B, the more LOW-leaning judge: 90%, 30%, 10%). On the 90-transcript held-out corpus (final rubric), majority-vote HIGH rate is 100% in every domain (Llama and Mistral predict HIGH on essentially every transcript in both corpora; Qwen 2.5 7B is again the only judge with substantial LOW-rate mass: 3%, 37%, 63% HIGH by domain under the final rubric). We report these figures because they bear directly on how several results in this paper should be read—most importantly Section 6.3’s judge-replacement test, where Llama and Mistral’s near-total mutual agreement on Career & Self-Improvement leaves little room for a third judge to influence the panel’s majority-vote output (Section 6.3 quantifies this directly).

Channel coverage. The 60-transcript evaluation corpus spans 46 unique channels (up to 5 transcripts from a single channel); the 90-transcript training-supplement corpus (Section 5.3) spans 50 unique channels (up to 7 from a single channel). The two corpora share 6 channels in common. Consequently, “held-out” in Section 5.3 means *document*-held-out (no transcript appears in both corpora), not *source*-held-out—see Limitations.

Initial rubrics. Three domain-specific rubrics at version 0, each specifying HIGH-signal criteria (methodological coherence, specificity, informational novelty, solution orientation) and LOW-signal patterns (generic advice, repetition, fear-based framing, promotional content, logical fallacies) with domain-appropriate threshold language.

Target threshold. $\tau = 0.80$. This is a modest margin above the metric’s chance level of ≈ 0.75 under independent judges (Eq. 1), not a claim that 0.80 represents strong agreement. This ≈ 0.75 chance level is derived entirely from CRUCIBLE’s own binary weighted- A metric and local judge panel; it is unrelated to, and should not be blended with, the separate 3-category pilot discussed in the Introduction and Appendix B, which concerns a different experiment in a different label space. We treat τ as a practical stopping rule for the calibration loop—a minimum bar past which further iteration is not obviously worthwhile—rather than as the paper’s primary quality signal; the chance-corrected metrics (Fleiss’ κ , Gwet’s AC1) reported in Section 5 are what we rely on to judge whether agreement reflects genuine convergence.

Coordinator retry. Up to 3 retries per call on JSON parsing failure; 5-second delay between attempts.

Runtime. 17–20 minutes per iteration (columnar order). 170.7 minutes total; the run terminated at the maximum-iteration cap of 10 iterations. Consumer MacBook (Apple Silicon), no discrete GPU; exact CPU/GPU offload configuration for the Ollama-served judge models was not recorded, and Ollama’s default behavior on Apple Silicon is to use Metal GPU acceleration where available, so “CPU-only” should not be assumed.

Baseline. CRUCIBLE’s own domain-specific v0 rubrics are the calibration starting point (Table 1). The earlier, differently-scoped pilot mentioned in the Introduction (frontier chat-model ensemble, domain-agnostic rubric, 3-category label space) motivated this work but is not a directly comparable quantitative baseline for the figures below; see Appendix B.

5 Results

5.1 Per-Domain Convergence

Table 1: Calibration results per domain.

Domain	Initial A	Best A	All-agree	Fleiss κ	Gwet AC1	Rubric ver.	Status
General Education	96.7%	96.7%	90.0%	−0.034	+0.929	v0 (0 changes)	Converged, iter. 1
Technology & AI	76.7%	81.7%	45.0%	+0.246	+0.286	v1 (1 change)	Converged, iter. 2
Career & Self-Impr.	68.4%	70.0%	10.0%	−0.350	−0.080	v1 (best of 2)	Stagnated

General Education converged immediately in iteration 1 with no rubric revision. Educational content with clear structural markers (lecture format, sequential explanation, named frameworks) produced consistent judgments across all models.

Technology & AI required one revision. The coordinator diagnosed that Qwen 2.5 7B was reading HIGH criterion 1 too narrowly—requiring production-system experience where the rubric intended methodological grounding. The revised criterion made the threshold explicit; agreement rose from 76.7% to 81.7%, crossing the 80% target in one step.

Career & Self-Improvement required two revision attempts: the first (v1) reached the best agreement of 70.0%, the second (v2) regressed to 68.4% and was discarded, so v1 was retained. Agreement never approached the 80% target. The coordinator’s diagnosis across iterations 1 and 2 consistently identified the same failure: Qwen 2.5 7B applied LOW criterion 1 (generic motivational language) to any transcript containing motivational phrasing, regardless of whether procedural content co-occurred. Revisions made the co-occurrence condition progressively more explicit, but the model continued misapplying the criterion. The coordinator flagged this as a model capability ceiling:

the model was failing to apply the co-occurrence condition consistently across long transcripts—a limitation at the model level, not a rubric ambiguity.

We report Fleiss’ κ and Gwet’s AC1 alongside raw weighted agreement, because raw agreement is inflated by label prevalence. The three domains separate cleanly under chance correction. Technology & AI shows fair agreement ($\kappa = +0.25$, AC1 = +0.286)—the only domain where a rubric revision carried agreement to the convergence target. Career & Self-Improvement shows below-chance agreement ($\kappa = -0.35$, AC1 = -0.080): the judges are systematically anti-correlated relative to that metric’s marginal-based chance expectation, which no rubric revision was able to change—consistent with, though not independent proof of, the capability-ceiling diagnosis discussed further in Section 6.3. General Education shows $\kappa \approx 0$ despite 96.7% raw agreement; this is the known prevalence paradox of κ under skewed marginals [19] (the domain is roughly 97% one class), and Gwet’s AC1 = +0.929 is less affected by the prevalence paradox under these marginals than raw agreement or Fleiss’ κ . The practical point is that CRUCIBLE’s contribution is not raising raw agreement per se—raw agreement is base-rate-dependent—but diagnosing which regime a domain occupies: rubric-fixable, capability-bounded, or prevalence-dominated.

5.1.1 The Stopping Rule Is Satisfied Before Agreement Is

The chance-corrected figures above expose a property of the convergence criterion itself, and it is the paper’s central finding. Equation 1 assigns weight 0.667 to every non-unanimous item, so A has a floor of 0.667 when no item is unanimous. Solving $A \geq \tau$ for the unanimous count u over N items gives

$$u \geq \frac{(\tau - 0.667) N}{0.333}, \quad (2)$$

which at $\tau = 0.80$ and $N = 20$ is $u \geq 8$. The convergence criterion is therefore met when 40% of items are unanimous and 60% still carry judge disagreement. Technology & AI crossed τ with $u = 9$ of 20; General Education, which converged without revision, had $u = 18$. Both are reported identically by the loop as “converged,” yet they differ by a factor of two in unanimity and by 0.643 in chance-corrected agreement (0.929 vs. 0.286).

This is a property of the threshold, not of the domains. The general form is that for any agreement statistic with floor f thresholded at τ , the required unanimity is $(\tau - f)/(1 - f)$ —so the more compressed the statistic’s range, the less agreement a given threshold demands. The practice this bears on is wider than our metric: thresholding raw percent agreement on binary labels carries a chance floor near 0.5, so a conventional 0.80 cutoff corresponds to roughly 60% unanimity and modest chance-corrected agreement.

We stated when defining Equation 1 that A is bounded below by 0.667 and that its chance level under independent judges is approximately 0.75. What we did not then do is carry that scale into the stopping rule, and the consequence is the result above: a domain can be certified as converged at a chance-corrected agreement of 0.286. For a safety-critical evaluation pipeline this is the risk-relevant point—an operator reading 81.7% agreement as convergence is reading a number whose floor is 66.7%. The remedy is arithmetic rather than procedural: report the floor alongside the threshold, or set the threshold on a chance-corrected statistic whose zero point is interpretable.

Initialization vs. iteration. We separate how much each domain’s result owes to writing a domain-specific starting rubric (v0, iteration 1) versus the iterative revision loop that follows.³ In

³Earlier drafts of this analysis expressed iteration 1’s contribution as a percentage of a gain measured from a 53% “domain-agnostic baseline.” That baseline comes from a different experiment (a frontier chat-model ensemble on a

General Education, initialization alone (iteration 1) achieved 96.7%—identical to the final result; the revision loop contributed nothing because none was needed. In Technology & AI, the domain-specific v0 rubric started at 76.7%, and one revision then added 5.0 points to cross threshold at 81.7%. In Career & Self-Improvement, the domain-specific v0 rubric started at 68.4%; one revision (v1) added a further 1.6 points to 70.0% before a second revision (v2) regressed performance and was discarded. Across all three domains, the domain-specific v0 rubric alone accounts for most of each domain’s final agreement level; the iterative loop’s marginal contribution is small in absolute terms, necessary to cross threshold in Technology & AI, and attempted-but-unhelpful in Career & Self-Improvement—itself informative, as it is what let us distinguish a rubric-fixable domain from one exhibiting the capability-ceiling pattern investigated further in Section 5.8.3 and directly tested via judge replacement in Section 6.3.

5.2 Convergence Trajectory

Table 2: Weighted agreement (%) per domain across iterations. ✓ = converged; △ = no improvement over best. Career’s second revision (v2) regressed at iteration 3 and was not retained; iterations 4–10 show no further improvement. The retained output is the best rubric, v1 (70.0%, iteration 2).

Iter.	Career & SI	Tech & AI	Gen. Edu.
1	68.4	76.7	96.7 ✓
2	70.0	81.7 ✓	96.7 ✓
3	68.4 △	81.7 ✓	96.7 ✓
4–10	68.4 △	81.7 ✓	96.7 ✓

Both converged domains remained stable across all subsequent iterations; regression checks confirmed no drift. For Career & Self-Improvement, the second revision (v2) regressed to 68.4% at iteration 3 and was rejected; no later revision exceeded the best observed rubric, so the retained system output is v1 at 70.0% from iteration 2. Agreement never reached the 80% target.

5.3 Held-Out Rubric Evaluation

Table 1 and Table 2 report agreement on the same 60-transcript corpus the coordinator revised rubrics against—so they cannot by themselves distinguish genuine generalization from fitting to that specific corpus. We address this with a single inference pass (no further coordinator revision) of the v0 and final calibrated rubrics over a 90-transcript corpus (30/domain) drawn from the training-supplement collection described in Section 4. We call this the *unseen-to-coordinator* set rather than a “held-out test set”: the strongest claim we can make with confidence, based on the documented project workflow rather than a logged, timestamped audit trail, is that this corpus was not shown to the coordinator during rubric revision. As established in Section 4, it is document-disjoint from the calibration corpus but shares 6 channels with it, so this is not a source-held-out test (see Limitations).

This shows the revision’s effect on inter-judge agreement is not confined to the calibration corpus. Career & Self-Improvement’s interval excludes zero; Technology & AI’s is borderline (lower bound at exactly 0); General Education’s rubric was never revised, so no change is possible or observed by

3-category label space; see Appendix B) and is not in the same units as the local-panel, binary weighted- A figures below, so we no longer report a percentage-of-gain figure against it.

Table 3: Unseen-to-coordinator evaluation (n=30/domain): weighted agreement under the v0 vs. final calibrated rubric, single inference pass, no further revision. CI is a 95% paired bootstrap interval (10,000 resamples) on the per-transcript weight difference (final minus v0).

Domain	v0	Final	Diff.	95% CI
Career & Self-Impr.	72.3%	78.9%	+6.7 pts	[+2.2, +11.1]
Technology & AI	84.5%	87.8%	+3.3 pts	[+0.0, +6.7]
General Education	98.9%	98.9%	+0.0 pts	[+0.0, +0.0]

construction. We report this as evidence the revision generalizes in its ability to *change inter-judge agreement* on unseen transcripts. It does **not** establish that the final rubric is more accurate or more human-aligned on this set—no human labels exist for it—and we do not describe this result as held-out validation of label quality anywhere in this paper.

The chance-corrected picture qualifies this further. Fleiss’ κ stays negative on this held-out set for both revised domains (Career: $-0.385 \rightarrow -0.268$; Technology & AI: $-0.184 \rightarrow -0.139$), the same prevalence-paradox pattern documented in Section 5 for the calibration corpus; Gwet’s AC1 is the metric that actually moves (Career: $0.072 \rightarrow 0.367$; Technology & AI: $0.578 \rightarrow 0.689$) and is the more informative read here given κ ’s known instability under this label distribution. Per-judge label prevalence explains why: on this corpus, Llama 3.1 8B and Mistral 7B predict HIGH on essentially 100% of transcripts in both domains, under *both* rubric versions—they do not move. Qwen 2.5 7B is the only judge whose behavior shifts (Career: $17\% \rightarrow 37\%$ HIGH rate; Technology & AI: $53\% \rightarrow 63\%$ HIGH rate), and this shift is what drives the reported agreement improvement. This is Qwen’s HIGH-rate converging toward the other two, effectively static, judges—not three-way convergence toward more nuanced shared judgment. Because Llama and Mistral already agree with each other on nearly every transcript in both domains, their majority vote is mechanically fixed regardless of Qwen’s individual prediction: the revision shifted the dissenting judge’s individual predictions but changed the ensemble’s *majority-vote annotation* on **0 of the 90** held-out transcripts (0 of 30 in each domain) between the v0 and final rubric. The weighted-agreement improvement reported above reflects Qwen’s predictions moving closer to unanimity with an already-fixed majority, not a change in which label the panel actually output on any transcript. Full per-judge prevalence and provenance detail are in Appendix D.

5.4 Downstream Validation: SNR Classifier Performance

We report this as a downstream *case study* in whether the calibrated labels are usable at scale, not as an independent validation of their agreement with the available human reference. The calibrated rubrics were used to drive labeling for a lightweight SVM classifier on the SNR binary classification task. Real transcripts labeled using CRUCIBLE-calibrated rubrics (applied by a separate frontier ensemble, per the companion paper [24]) were combined with 300 synthetically generated transcripts for class balance, and the classifier was evaluated on a held-out set of 60 real transcripts.

Model	F1 (HIGH signal)	Recall (HIGH signal)
SVM on rubric-calibrated labels (mixed real+synthetic)	0.871	0.964
SVM on synthetic-only training (companion paper)	0.747	—
Majority-class baseline	0.000	—

We did not train a classifier on labels from the uncalibrated pilot pipeline (Introduction); the

majority-class baseline ($F1 = 0.000$) indicates the task is non-trivial. The gap between the synthetic-only ablation ($F1 = 0.747$, reported in the companion paper [24]) and the mixed real+synthetic result ($F1 = 0.871$) indicates the real, calibrated-rubric-labeled transcripts contribute materially beyond what synthetic data alone provides—though, per Limitations, this does not fully rule out the mixed result partly reflecting the classifier learning to separate synthetic-generation artifacts rather than purely calibrated-label quality. Because the held-out labels were produced by the same silver-labeling methodology rather than independent human-validated labels, this $F1$ reflects internal consistency with the labeling pipeline rather than external, human-validated accuracy—Section 5.7 addresses this directly. Full experimental details of the SNR classifier—dataset construction, synthetic augmentation strategy, and complete evaluation protocol—are reported in the companion paper [24] (see footnote 1).

Train/test split and label independence. The classifier is trained on 389 transcripts: 89 real transcripts with silver labels assigned via the same calibrated-rubric ensemble methodology, combined with 300 synthetic transcripts (150 HIGH, 150 LOW, balanced across domains) generated from structured prompts with labels assigned by construction, intended to match the generating prompt’s target class. Evaluation uses a disjoint, held-out set of 60 real transcripts—the same 60-transcript corpus used throughout this paper—selected and labeled prior to any classifier training. The 89 training-supplement transcripts and 300 synthetic transcripts are disjoint from the 60-transcript evaluation set by construction (no transcript appears in both); synthetic transcripts correspond to no real video, and the 89 real training-supplement transcripts are document-disjoint from evaluation but—per Section 4—share some channels with it (6 of 46 evaluation-set channels also appear in the 90-transcript training-supplement pool from which the 89 are drawn), so this is document-held-out, not source-held-out. We note directly, per the concern motivating Section 5.6, that the evaluation labels are *not* independent of the calibration process in the sense of an external human reference—they are assigned by the same class of frontier-ensemble judges used for the training supplement, applying the calibrated rubric. This is precisely the circularity Section 5.6 is designed to address. That section’s conclusion is a caveat on the $F1$ reported above, but a weaker one than we previously claimed: human–system match over the full 20-per-domain sample is 75%, which is *below* the 83% majority-class baseline, so the match statistic neither supports nor impugns the evaluation labels here. What it does establish is that no external human check validates these labels, and the earlier per-domain figures we quoted in this role (90%/50%/50% at $n = 10$) are prevalence-dominated and retracted. The $F1$ score demonstrates the calibrated labels are *learnable* by a downstream classifier; it does not by itself demonstrate the labels are *correct*.

5.5 Irreducible Cases

The coordinator flagged two Technology & AI transcripts as irreducible—genuine content ambiguity rather than rubric failures—and excluded them from stagnation counting. We were unable to independently verify these specific cases against surviving logs (Appendix E explains why and reports a sensitivity check instead), so we do not name the specific transcripts here as the original draft of this paper did.

5.6 Human Validation

A known limitation of LLM-as-judge pipelines is that inter-judge agreement among automated annotators may reflect shared model bias rather than agreement with independent human judgment, and without a human reference point, convergence cannot be distinguished from consensus error. We address this directly with an independent human validation study.

5.6.1 Design

The study covers the entire corpus: 20 transcripts per domain, run in two rounds of 10. Each round recruited three independent annotators per domain via Prolific [21], 18 annotators in total, with no annotator contributing to more than one domain or more than one round. Round 1 scored the even-indexed transcripts within each domain block and round 2 the odd-indexed remainder, so the two rounds partition the corpus exactly. Each round also carried one hidden attention-check item per domain, not included in the analysis below. Annotators were required to be proficient in English and hold a Prolific approval rating above 95%. Annotators were shown only the raw transcript text—not the calibrated rubric, not the domain label, and not CRUCIBLE’s output—and asked to judge, in their own words, whether each transcript was substantive and valuable or generic and low-value, with a mandatory written justification. This is an unaided-judgment design only; we did not run a second condition showing annotators the calibrated rubric (see Limitations for what this does and does not establish). This design is deliberate: showing annotators the calibrated rubric would test whether humans can *follow* CRUCIBLE’s criteria, not whether CRUCIBLE’s criteria independently match human judgment. Quality was enforced via a hidden attention-check item per domain (adversarial or unambiguous content with a known expected label), a mandatory content-grounded justification, and task instructions designed to flag fast or low-effort responses for rejection. One of nine submissions (Technology & AI domain) was rejected and replaced, on three converging factors confirmed by reviewing the full submission: a completion time of 39:06 against this batch’s displayed median of 140:00, failure of the domain’s attention-check item, and justifications consisting largely of copy-pasted transcript excerpts rather than substantive reasoning. This was a multi-factor judgment made after reviewing the submission’s full content, not a blind, mechanically-triggered rejection; we report a sensitivity check in Appendix F showing the domain-level results are unchanged whether this submission or its replacement is used.

5.6.2 Results

We report three quantities per domain: (1) inter-annotator agreement among the three humans in each round, via Gwet’s AC1 [19], with 95% percentile bootstrap intervals over items (10,000 resamples) on the pooled 20; (2) the agreement between human majority-vote consensus and CRUCIBLE’s calibrated label; and (3) the *majority-class baseline* for that domain-round—the score a rater achieves by assigning every item the domain’s dominant system-label class. The third quantity is essential and was absent from our earlier report: where the reference labels are heavily skewed, a high match rate is attainable without any discriminative judgment at all, and the match statistic must be read against that baseline rather than against chance. Channel identity is recorded for this corpus (Section 4); Appendix G reports these same intervals resampled by channel rather than by transcript, since several channels contribute more than one transcript to this 30-item sample.

Two results follow, and they point in different directions.

A correspondence between judge and human reliability. Chance-corrected agreement among the human annotators ranks the three domains in exactly the order the judge panel does: General Education highest (human AC1 = 0.70 pooled, judge AC1 = 0.929), Technology & AI second (0.33 vs. 0.286), Career & Self-Improvement lowest (0.24 vs. −0.080). With three domains this is a rank statement rather than a tested relationship, but the magnitudes agree closely in the two harder domains. The judges find difficult what the humans find difficult. This is evidence against reading the judge panel’s failures as arbitrary, and it is the reason we relocate the paper’s central claim from human misalignment to the stopping rule.

Table 4: Human validation at 20 transcripts per domain, scored by two disjoint cohorts of three annotators (round 1 / round 2). AC1 is Gwet’s AC1 among the three annotators within a domain-round; the pooled column gives AC1 over all 20 items with a 95% percentile bootstrap interval (10,000 resamples over items). “vs. CRUC.” is the share of transcripts where human majority-vote consensus matches CRUCIBLE’s calibrated label. “Base” is the majority-class baseline: the score a rater achieves by labelling every item with the domain’s dominant system-label class. Bold marks the only cell that exceeds its own baseline.

Domain	AC1 r1 / r2	AC1 pooled (95% CI)	vs. CRUC. r1 / r2	Base
General Education	0.63 / 0.76	0.70 [0.43, 0.89]	90% / 100%	100%
Technology & AI	0.63 / -0.03	0.33 [-0.02, 0.64]	50% / 80%	60%
Career & Self-Impr.	0.33 / 0.23	0.24 [-0.06, 0.56]	50% / 80%	90%
Overall	—	—	75%	83%

A negative result on human–system match. The match statistic itself carries almost no evidential weight in this design. Five of the six domain-round cells fail to beat a rater who simply assigns the domain’s dominant class to every item, and the pooled rate of 75% sits below the corresponding 83% baseline. General Education’s reference labels are 10/10 HIGH, so its 90% and 100% match rates are tied or beaten by constant labelling and cannot constitute evidence of alignment in either direction. Only Technology & AI round 2 (80% against a 60% baseline) exceeds its own baseline, on ten items.

What the enlarged sample does and does not establish. An earlier version of this study, at $n = 10$ per domain, reported a between-domain contrast in human–system match (9/10 for General Education vs. 5/10 for Technology & AI) as preliminary evidence of an alignment dissociation. Doubling to $n = 20$ did not confirm that contrast; it removed its basis. Technology & AI’s second, independent cohort matched the system on 8 of 10 rather than 5 of 10, and the two General Education cohorts are tied or beaten by constant labelling in both rounds. We report this as a failed replication of our own earlier result and retract the dissociation claim.

On the reliability side we state precisely what clears zero. Among the judges, General Education exceeds Technology & AI (+0.64, 95% CI [+0.31, +0.94]) and Career (+1.01, [+0.73, +1.21]); Technology & AI versus Career is marginal (+0.37, [−0.01, +0.74]). Among the humans, only General Education versus Career clears zero (+0.46, [+0.04, +0.80]); General Education versus Technology & AI does not (+0.37, [−0.04, +0.77]), and Technology & AI versus Career is indistinguishable (+0.09, [−0.39, +0.52]). Technology & AI’s human AC1 does change between cohorts (+0.66, [+0.03, +1.08]), so its human reliability is cohort-dependent where General Education’s is not. At 20 items and three raters per cohort these intervals are wide; we present the estimates as a ranking with stated uncertainty, not as resolved effect sizes.

Two design caveats apply throughout. Each domain-round was scored by a distinct, non-overlapping set of three annotators, so domain effects and annotator-cohort effects cannot be fully separated; the two rounds within a domain give two independent estimates of that domain’s reliability rather than one, which is what makes the cohort-dependence statement above possible. And because the cohorts are disjoint, chance-corrected agreement is computed per round and only then pooled over items; a single agreement figure computed across non-overlapping rater sets would conflate between-cohort with within-cohort disagreement.

5.6.3 Implications

These results require us to revise the framing of CRUCIBLE’s central claim, and in a different direction from the one our earlier report proposed. The human data do not show the judges to be misaligned with a well-defined human standard; they show judge and human reliability tracking each other across domains. What does not survive contact with the enlarged sample is the convergence *criterion*: the domain the loop certified as converged at 81.7% has a chance-corrected inter-judge agreement of 0.286, and the threshold it crossed is satisfied while 60% of items still carry disagreement (Section 5.1.1). Automated inter-judge convergence is not sufficient evidence of alignment with human judgment—but the mechanism is an agreement statistic whose scale was never established, not a divergence between the panel and its human reference. We therefore do not claim CRUCIBLE eliminates the need for human oversight; we show instead that periodic, targeted human validation—at the scale used here, a small stratified sample per domain—is advisable, and appears necessary before treating automated agreement alone as evidence of validity in settings similar to this one, to determine *which* domains’ automated convergence can be trusted, a determination the convergence metric alone cannot make.

We did not recalibrate CRUCIBLE’s rubrics toward the human labels obtained here. Doing so would reintroduce exactly the circularity this validation study is designed to break: a rubric tuned to match a specific human sample provides no evidence of generalization beyond that sample. The role of this validation is diagnostic, not corrective.

5.6.4 Ethics Statement (Human Subjects)

The human validation study involved data collection from human participants via the Prolific platform. No formal institutional ethics review or exemption was sought for this study; participation was governed solely by Prolific’s standard platform consent and terms. Participants were compensated at \$12.69/hour, above Prolific’s platform minimum, for a task with a 140-minute estimated completion window. Participation was voluntary; no personally identifiable information was collected beyond participants’ Prolific IDs, which are pseudonymous and not linked to real-world identity in our data. Task descriptions accurately represented the nature of the work (judging transcript content quality) with no deceptive framing. Data was collected and analyzed in aggregate; no individual participant responses are reported in a way that could identify a specific participant. One submission (Technology & AI domain) was rejected and replaced during data collection; see Section 5.6’s Design subsection for the confirmed rejection criteria and Appendix F for the resulting sensitivity check.

5.7 Does the Frontier Ensemble That Labeled the Classifier Track Human Judgment?

Section 5.6 validates the *local calibration panel* that CRUCIBLE revises rubrics against. It does not by itself say anything about the separate, stronger frontier ensemble (GPT-4o mini, Gemini 2.5 Flash, Llama 3.3 70B) that actually produced the labels used to train and evaluate the downstream SNR classifier (Section 5). We close this gap directly using the round-1 30-item human reference set, since this frontier ensemble’s per-transcript labels for the original 60-transcript corpus were already available. This comparison was not re-run against the enlarged 60-item reference, and it inherits the prevalence limitation established in Section 5.6: match against human consensus is not a measurement of alignment in this design. We retain it because the *relative* statement it supports—that the frontier ensemble tracks human judgment no more closely than the local panel does—is a within-comparison contrast on identical items, and does not rest on either figure being

an alignment estimate.

Table 5: Frontier ensemble vs. human majority consensus ($n = 30$, 10/domain). Gemini 2.5 Flash returned a valid label on 20/30 items; majority vote is robust to this since 2 of 3 judges suffice.

Comparison	Agreement	Cohen’s κ	Gwet’s AC1
Frontier ensemble majority vs. human	56.7%	0.133	0.178
GPT-4o mini vs. human	56.7%	0.184	0.157
Gemini 2.5 Flash vs. human ($n = 20$)	55.0%	0.043	0.153
Llama 3.3 70B (Groq) vs. human	60.0%	0.174	0.253
Local calibration panel vs. frontier ensemble	66.7%	0.333	0.400

The production frontier ensemble that actually generated the SNR classifier’s training and evaluation labels tracks human judgment about as loosely as the local calibration panel does—no individual frontier judge exceeds 60% agreement with human consensus, and the ensemble majority (56.7%) is comparable to the local panel’s own 63.3% (Table 4). This is not a minor footnote-level caveat: it is direct evidence that the downstream classifier’s $F1 = 0.871$ result (Section 5, reported there as a usability case study, not a claim about agreement with the available human reference) rests on labels of similar human-alignment reliability to the ones scrutinized throughout this paper, not on some independently more trustworthy labeling process. Local-panel and frontier-ensemble labels agree with *each other* more (66.7%) than either agrees with humans—consistent with this paper’s broader finding that automated agreement, even between differently-sourced judge ensembles, does not imply agreement with the available human reference.

Missing-label sensitivity check. Gemini 2.5 Flash failed to return a valid label on 10 of the 30 human-validation items; these 10 items are excluded from Gemini’s own individual-agreement row in Table 5 only, and the ensemble-majority row above uses whichever 2 or 3 of the 3 judges returned a valid label on each item (majority requires only 2 of 3, so this is well-defined throughout). On the 10 items where Gemini’s label is missing, the remaining two judges (GPT-4o mini and Llama 3.3 70B) agree with each other on 9 of 10 and disagree on 1 of 10, so Gemini’s absence rarely leaves the majority ambiguous. As a direct check, restricting to the 20 items where all three frontier judges returned a valid label (dropping the 2-of-3 convenience entirely) gives frontier-majority-vs-human agreement of 55.0% (Cohen’s $\kappa = 0.100$, Gwet’s AC1 = 0.120)—close to the full-sample, 2-of-3-robust figure of 56.7% ($\kappa = 0.133$, AC1 = 0.178) reported above, so the missing-label handling does not materially change the finding.

5.8 Rubric Portability Across Judge Models

A natural question is why a single frontier model, applied zero-shot, would not suffice in place of the full calibration loop. We test this directly, and in doing so uncover a second finding: the calibrated rubric does not transfer cleanly to a judge model different from the one it was calibrated on.

5.8.1 Frontier Baseline Design

We compare three approaches against the same 30-transcript human ground truth used in Section 5.6: (1) the local three-judge panel applying its calibrated rubric (the system described in this paper); (2) GPT-5.6 Sol [23] (OpenAI, general availability July 9, 2026) applying a naive zero-shot prompt with no rubric; and (3) GPT-5.6 Sol applying the same calibrated rubric used by the local panel. All GPT-5.6 Sol calls for this experiment were made on July 19, 2026, ten days after general

availability. We use a non-Anthropic model for this baseline because Claude serves exclusively as the calibration coordinator throughout this work and was not evaluated as a judge, to avoid conflating baseline performance with the model family responsible for rubric design. GPT-5.6 Sol does not support temperature=0 at time of writing; results reflect the model’s default sampling temperature.

Table 6: Accuracy against independent human judgment (n = 30, 10 per domain), comparing the local calibrated panel against a frontier model with and without the calibrated rubric.

Approach	Career	Tech & AI	Gen. Ed.	Overall
Local panel + calibrated rubric	50%	50%	90%	63.3%
GPT-5.6 Sol, no rubric	60%	30%	100%	63.3%
GPT-5.6 Sol, + calibrated rubric	20%	40%	90%	50.0%

5.8.2 Findings

Overall, the local calibrated panel and a single zero-shot frontier pass achieve identical accuracy against human judgment (63.3%)—the 10-iteration calibration loop does not categorically outperform a single frontier zero-shot pass in aggregate. However, giving the frontier model the *same* calibrated rubric text does not add value, and in Career & Self-Improvement, degrades accuracy substantially (60% → 20% against human judgment). Comparing the same frontier model’s outputs against the local panel’s labels directly (rather than against human judgment) shows the rubric *does* make GPT-5.6 Sol’s outputs converge toward the local panel’s specific behavior (agreement with the local panel rises from 0.500 to 0.633 when the rubric is added). Taken together, this indicates the calibrated rubric teaches a frontier judge to mimic the local ensemble’s particular behavior pattern—including the local ensemble’s specific errors—rather than making the frontier judge more accurate against the available human reference. The rubric appears co-adapted to the specific quirks of the local judge ensemble it was calibrated on (Section 5, which documents Qwen 2.5 7B’s specific misreadings that motivated each revision), and this adaptation does not generalize to a different judge model.

We verified this baseline is not a single-draw artifact: repeating both the no-rubric and rubric-prompted GPT-5.6 Sol calls five times per transcript (600 total calls) shows 87–93% of transcripts receive an identical label across all five draws, and the majority-of-five accuracy against the available human reference matches the original single-draw result exactly for both variants.

5.8.3 Reasoning-Before-Label Ablation

Motivated by the question of whether Career & Self-Improvement’s stagnation (Table 1) reflects a single-pass prompting limitation rather than a true model capability ceiling, we added an intermediate reasoning step to the local judges’ prompts, in the style of chain-of-thought prompting [20] (reasoning generated before the label, not after), and re-scored all three domains against their existing calibrated rubrics, using the same three local models at temperature=0.0. We call this a *reasoning-before-label prompt* rather than chain-of-thought throughout, since we have no access to the models’ actual internal computation—only to text generated before the label in response to an instruction to reason first.

The reasoning step reliably raises inter-judge agreement in the two domains that had room to improve (Career: +8.3 points weighted; Technology & AI: +11.6 points, Gwet’s AC1 nearly triples),

Table 7: Reasoning-before-label ablation across all three domains: inter-judge weighted agreement and accuracy against the available human reference, without vs. with the reasoning step.

Domain	Inter-judge A		vs. Human	
	No reasoning	Reasoning	No reasoning	Reasoning
Career & Self-Impr.	0.684	0.767	0.500	0.400
Technology & AI	0.817	0.933	0.500	0.500
General Education	0.967	0.917	0.900	0.900

but it never *helps*, and in Career actively *hurts*, agreement with human judgment. Career’s human-agreement drops from 50% to 40% with the reasoning step (Cohen’s κ flips from +0.14 to −0.25); Technology & AI is flat on raw agreement but its AC1 drops; General Education, already near-saturated, is unchanged against humans and slightly dips on inter-judge agreement, consistent with a ceiling effect.

This result rules out simple single-pass prompting failure as the sole explanation for Career & Self-Improvement’s stagnation—the judges *can* be made to agree more with reasoning added—but shows that forcing reasoning-before-label makes the panel more internally consistent without making that consistency more correct. We revise our diagnosis accordingly. The enlarged human study (Section 5.6) shows Career’s own annotators agreeing with each other barely above chance (AC1 = 0.24 pooled, 95% CI [−0.06, 0.56]), the lowest of the three domains and matching the judge panel’s ordering. The original stagnation therefore reflects neither a simple inability of the judges to converge nor a misalignment against a well-defined human standard, but a domain in which the substantive/generic distinction is weakly determined for judges and humans alike—the capability-bounded cell of our own trichotomy, now with rater-side evidence. Explicit reasoning makes the panel more internally consistent without making it more correct. We used a single, untuned reasoning-before-label prompt design across all domains; whether a more carefully engineered prompt would alter this pattern is untested and left for future work.

5.8.4 Revision-Sensitivity Analysis

The findings above raise a further question: how much of the coordinator’s v1 revision’s effect (Technology & AI, 76.7% → 81.7%) is attributable to the coordinator’s specific diagnostic reasoning, versus simply having *any* reasonable revision informed by the disagreement evidence, versus having any revision at all? We call this a *revision-sensitivity analysis*, not an ablation: a single alternative revision per condition cannot isolate the coordinator’s causal contribution, only indicate whether the pattern is sensitive to how the revision is produced.

We regenerated the v0-iteration disagreement cases for Technology & AI’s 20 transcripts via a fresh deterministic pass (temperature = 0.0; these were not persisted from the original run, so this is how they are recovered) and generated two alternative revisions, both via GPT-5.6 Sol⁴: (a) *disagreement-informed*, seeing the same disagreement cases and criterion quotes the original coordinator saw, but not the coordinator’s own revision text; and (b) *rubric-only*, seeing only the v0 rubric with no disagreement evidence at all. Both alternatives were run through the same local three-judge panel on the same 20 transcripts, single inference pass, no further iteration.

⁴No Anthropic API access was available for this analysis; both alternatives substitute GPT-5.6 Sol for Claude Sonnet, the original coordinator model. This is a genuine confound: differences from the original coordinator’s v1 rubric may partly reflect model-family differences in rubric-writing style rather than purely the evidence-visibility manipulation this analysis targets.

Table 8: Revision-sensitivity analysis, Technology & AI (n=20). Regenerated v0 matches the originally reported 76.7% exactly, indicating no detectable model drift since the original run.

Condition	Weighted A	Fleiss κ	Gwet AC1
v0 baseline (regenerated)	0.767	0.067	0.067
Original coordinator v1 (Claude Sonnet)	0.817	0.246	0.287
Alt (a) disagreement-informed (GPT-5.6 Sol)	0.734	−0.111	−0.026
Alt (b) rubric-only (GPT-5.6 Sol)	0.767	0.063	0.071

In this single trial, the original coordinator outperformed both alternatives, and having disagreement evidence was not by itself sufficient: alt (a), given the *same* evidence the coordinator saw, performed *worse* than alt (b)’s blind, no-evidence revision, which in turn was statistically indistinguishable from not revising at all. Looking at the generated text, alt (a) rewrote multiple HIGH criteria broadly and more restrictively, where the original coordinator made a single surgical clause edit (Appendix A); the broad rewrite appears to have destabilized rather than fixed the panel. We do not read this as establishing that the coordinator’s diagnostic reasoning is necessary or superior in general: both alternatives were generated by a different model than the original coordinator, so this comparison cannot cleanly separate “the coordinator’s reasoning is better” from “Claude Sonnet writes better surgical rubric edits than GPT-5.6 Sol for this task.” We report it as a sensitivity finding, not a controlled ablation result.

These findings—rubric non-transfer across judge models, a reasoning-before-label step increasing agreement without increasing agreement with the available human reference, the frontier ensemble tracking human judgment about as loosely as the local panel (Section 5.7), and the Section 5.1.1 finding that the convergence threshold is met while most items still carry disagreement—share a common implication: **making automated judges agree more with each other is not the same as making them more correct.** Section 6.3 adds a further, more direct demonstration via judge replacement. Two scope notes apply. The rubric-portability and reasoning-before-label analyses in this section, and the frontier comparison in Section 5.7, all score against the original round-1 30-item human reference and were not re-run against the enlarged 60-item sample; they should be read as related diagnostic analyses on one reference set, not as independent replications, since a property specific to that set could affect several of them jointly. And because match against human consensus is prevalence-dominated in this design (Section 5.6), the human-agreement side of each comparison supports the *relative* statement that the metric moved while human agreement did not, rather than any absolute estimate of alignment.

6 Analysis and Discussion

Every result below is organized around the two questions posed in the Introduction: does disagreement-driven rubric revision generalize past the transcripts it was tuned on (Section 5.3), and does higher inter-judge agreement correspond to higher human alignment (Sections 5.6, 5.7, 5.8, and 6.3 below). Table 9 summarizes when each analysis was run relative to the frozen system, to make explicit that nothing downstream of the primary calibration run fed back into rubric or system design.

Table 9: Chronology of analyses. “Could influence design?” marks whether a stage’s output could, in principle, have fed back into rubric or system design. Only the first two stages could; every analysis from human validation onward was run against the frozen system and rubrics.

Stage	Data used	Could influence design?
Initial rubric design (v0)	Domain knowledge, no corpus data	Yes
Primary calibration run	60-transcript corpus, iter. 1–10	Yes
Human validation (§5.6)	Full corpus (20/domain, 2 rounds)	No
Frontier ensemble vs. human (§5.7)	Round-1 30-item subset	No
Rubric portability (§5.8)	Round-1 30-item subset	No
Reasoning-before-label (§5.8.3)	Round-1 30-item subset	No
Revision-sensitivity (§5.8.4)	Tech & AI subset (alt. rubrics, not adopted)	No
Held-out rubric evaluation (§5.3)	90-transcript unseen-to-coordinator set	No
Judge replacement (§6.3)	Career subset	No
Irreducible-case sensitivity (App. E)	Existing calibration output	No
Prolific rejection sensitivity (App. F)	Existing annotation records	No

6.1 What CRUCIBLE Addresses

CRUCIBLE substitutes structured inter-judge disagreement—measurable, diagnosable, and actionable—for the implicit and unauditable disagreement that characterizes single-annotator labeling.

This is not a claim that ensemble labels are unbiased. Each judge carries its own training biases. But the ensemble structure makes those biases visible: when one model systematically diverges from the other two, the coordinator can name the pattern. Human annotator bias is usually invisible unless multiple annotators are compared—precisely the expensive intervention this system replaces.

6.2 The Capability Ceiling Finding

The Career & Self-Improvement ceiling—a best of 70.0%, short of convergence—shows that a class of disagreement exists that rubric revision did not fix under this judge and prompt setup. Without the CRUCIBLE diagnostic loop, an operator seeing 70.0% would not know whether to keep revising or replace the judge. The stagnation pattern plus the coordinator’s consistent diagnosis across both revision attempts—attributing it to a model-level inability to track co-occurrence patterns across long documents—suggested the rubric was not the limiting factor. The natural prescription is judge replacement or ensemble augmentation, not further revision; Section 6.3 tests this directly, though as that section shows in detail, the specific test we were able to run cannot fully distinguish a genuine judge-level capability limitation from a structural property of this particular three-judge panel.

6.3 Judge Replacement: A Direct Test

Section 6.2 recommends judge replacement, not further rubric revision, once a domain is diagnosed as capability-bounded. We test this directly on Career & Self-Improvement, replacing Qwen 2.5 7B—the judge the coordinator’s diagnosis consistently implicated—with **granite3.1-dense:8b** (IBM, 8.2B parameters, 131,072-token context), keeping Llama 3.1 8B and Mistral 7B unchanged and the retained v1 rubric fixed. This is the only replacement model attempted: we selected it before running any evaluation, for comparable scale to Qwen 2.5 7B and a training lineage distinct from Meta, Mistral AI, and Alibaba—explicitly not Gemma 2 9B, which Section 3.5 already disqualified as degenerate. Trying multiple candidates and reporting the best-

performing one would constitute model selection on the same small human-reference set used to score it below, which we specifically avoided.

Table 10: Judge replacement on Career & Self-Improvement (n = 20 for inter-judge agreement; n = 10 for human alignment).

Panel	Inter-judge <i>A</i>	vs. Human	vs. Human (κ / AC1)
Original (Llama, Mistral, Qwen)	0.700	50.0%	0.138 / 0.083
Replacement (Llama, Mistral, Granite)	0.883	50.0%	0.138 / 0.083

Replacing the diagnosed-bottleneck judge raises inter-judge agreement substantially, from 0.700 to 0.883. Accuracy against the 10 human-labeled Career transcripts is *bit-for-bit identical* before and after: the replacement panel’s majority-vote label matches the original panel’s on every one of the 10 items, so agreement, Cohen’s κ , and Gwet’s AC1 against human judgment are unchanged to three decimal places.

Why: a mechanical check on the full 20-transcript set. We looked more closely at why, using the full 20-transcript Career set (not just the 10 with human labels), and found this dissociation is mechanically guaranteed rather than an independently interesting empirical outcome. Llama 3.1 8B and Mistral 7B—the two judges held fixed across both panels—already agree with each other on 18 of the 20 (90%) Career transcripts (Llama predicts HIGH on 20/20, Mistral on 18/20); on those 18, the majority-vote label is determined entirely by Llama and Mistral, regardless of what the third judge outputs. On the remaining 2 transcripts (Llama = HIGH, Mistral = LOW), both the original judge (Qwen 2.5 7B, 10% HIGH rate on this domain) and the replacement (granite3.1-dense, 65% HIGH rate) happened to vote LOW, so the majority stayed LOW under both panels there too. The result: the majority-vote label changed on **0 of the 20** Career transcripts between the original and replacement panel. The inter-judge agreement metric moved (0.700 $\rightarrow$ 0.883) because granite3.1-dense agrees with the Llama+Mistral majority far more often than Qwen did (65% vs. 10% HIGH rate), converting many 2-1 splits into 3-0 unanimity—not because the panel’s actual output changed on any transcript.

This means the finding is narrower than the table alone suggests: what we have directly demonstrated is that judge replacement changed whether the panel was unanimous or split, not what the panel’s actual annotations were, and by construction the annotations *could not* have changed on this particular panel composition. This result is *consistent with*—but does not independently establish—this paper’s broader claim that fixing inter-judge agreement does not fix agreement with human judgment: the majority-vote label here was mechanically unchangeable by any third judge given how strongly the other two already agreed, so the null result on human alignment cannot be attributed with confidence to *correcting* a genuine capability gap versus simply *being unable to move* the panel’s output at all. The label prevalence underlying this is informative too: both panels produce the same distribution of HIGH/LOW calls at the majority-vote level on this subset, which is consistent with the two panels reaching the same place by different within-panel routes rather than the replacement judge correcting Qwen’s specific misreadings—though, per the analysis above, “reaching the same place” was close to inevitable given the fixed pair’s own agreement rate.

We read this cautiously in two respects: it is a single replacement-model trial on one domain (see Limitations), and, as detailed above, the extent to which any third judge could have influenced the majority-vote outcome at all is itself limited on this panel. This experiment is better read as evidence that judge replacement can convert a split panel into a unanimous one without changing its output, rather than as a demonstration that judge replacement is a general recovery strategy

for capability-bounded disagreement (see Limitations for a fuller discussion of this constraint).

6.4 Degenerate Judge Detection

A judge assigning a single label to all documents makes agreement appear high when it is driven by shared degeneracy rather than shared understanding. We treat discriminative validity as a pre-screening gate: no model enters a judge panel without demonstrating a non-degenerate label distribution across a diagnostic sample.

6.5 Output-Stability Recheck for Converged Domains

Once a domain converges, we re-evaluate it with the same rubric and the same three judge models on the iteration immediately following convergence, checking whether the weighted agreement rate holds. Each coordinator call is a fresh, stateless API request scoped to a single domain (it receives only that domain’s rubric, disagreement cases, and revision history; no information about other domains’ prompts, revisions, or outputs is ever included), so this check is not, and cannot be, a defense against one domain’s revision “priming” the coordinator’s phrasing on another domain via shared state—no such state exists in this implementation. What it does test is judge-level output stability: whether the same rubric applied to the same transcripts by the same three models reproduces the same agreement rate on a second pass, since a frontier API coordinator’s own non-determinism (Limitations) is a property of the coordinator, not the judges, but serving-side variance in the local judge models is not fully ruled out either. No regression (agreement drop > 10 percentage points) was observed in the run reported here across any of the domains checked.

7 Limitations

Single experiment. Results are for 60 transcripts across three domains. Generalization requires validation on other domains and document types.

Non-standard agreement metric. The statistic the loop optimizes (Eq. 1) is not a standard chance-corrected measure, and its floor of 0.667 is what makes the $\tau = 0.80$ threshold permissive (Section 5.1.1). We report Fleiss’ κ and Gwet’s AC1 alongside it throughout, but the *optimization* and *stopping* decisions in the reported run were driven by the uncorrected statistic, so the released rubrics are products of that criterion. Re-running the calibration loop with the threshold set on AC1 directly would test whether the same rubric revisions are selected; we have not done this, and it is the most direct follow-up this finding implies.

Two-cohort pooling. Each domain’s 20 items were scored by two disjoint cohorts of three annotators. This yields two independent reliability estimates per domain rather than one, which is what supports the cohort-dependence statement in Section 5.6, but it also means no annotator scored both halves. Chance-corrected agreement is therefore computed within round and pooled over items; a single figure computed across non-overlapping rater sets would conflate between-cohort with within-cohort disagreement. Three raters per cohort remains the binding constraint on the width of every interval we report.

Model-specific ceiling. The Career stagnation was diagnosed as specific to Qwen 2.5 7B under our prompt and setup; we cannot rule out that a different prompt formulation or judge model would shift the ceiling. Future work should evaluate judge replacement and prompt variation as recovery strategies. This attribution rests on a single, untuned prompt template per judge model used throughout the calibration run reported here; prompt reformulation remains untested as an

alternative explanation for the ceiling and should be evaluated alongside judge replacement before concluding the ceiling is inherent to the model rather than an artifact of this specific prompt.

Judge replacement: tested once, narrowly. Section 6.3 now tests Section 6.2’s recommendation directly: replacing Qwen 2.5 7B with granite3.1-dense:8b on Career & Self-Improvement raised inter-judge agreement substantially ($0.700 \rightarrow 0.883$) while leaving accuracy against the 10 human-labeled transcripts completely unchanged. This is a single replacement-model trial on a single domain, with only one candidate model attempted; it does not establish that judge replacement is a general recovery strategy.

Judge replacement’s strength depends on how much the third judge could actually influence the outcome. As Section 6.3 details, the two retained judges (Llama, Mistral) already agreed with each other on 90% of this domain’s 20 transcripts, and on the remaining 2 both candidate third judges happened to vote the same way; the majority-vote label therefore changed on 0 of 20 transcripts between the original and replacement panel. This means the experiment, as run, could only ever have shown a dissociation between inter-judge agreement and human alignment on this domain—it was structurally incapable of showing the opposite, because the panel’s output could not move. The human-alignment half of that comparison is additionally prevalence-dominated (Section 5.6), so the unchanged human-agreement figure carries little information either way. A domain or panel where the two retained judges disagree with each other more often (so the third judge’s identity genuinely determines more majority votes) would be a stronger, more informative test of whether judge replacement recovers human alignment along with inter-judge agreement; we did not have such a case available in this run’s data. Readers should weight this result accordingly: it demonstrates a real property of the diagnostic loop (replacement resolved the agreement symptom) but the null result on human alignment is consistent with, rather than independent confirmation of, this paper’s broader claim.

Silver labels. Calibrated labels retain the biases of the three judge models. Section 5.6’s human validation study now covers the full corpus at 20 transcripts per domain, but it addresses this less than we previously claimed: human–system match is 75% pooled against an 83% majority-class baseline, and five of six domain-round cells fail to beat constant labelling, so the match statistic does not function as a check on label quality in this design. The study’s informative output is the reliability comparison, not the match rate. It remains an unaided-judgment sample with three raters per cohort, not a large-scale or rubric-guided gold standard. For high-stakes tasks, human adjudication of remaining disagreements at greater scale is recommended.

Synthetic data in classifier training. Of the 389 transcripts used to train the downstream SVM classifier (Section 5), 300 (77%) are synthetically generated with labels assigned by construction, intended to match the generating prompt’s target class, rather than assigned by the calibrated-rubric ensemble. The reported $F1=0.871$ may therefore partly reflect the classifier separating synthetic-generation artifacts from real transcripts, rather than purely the quality of the calibrated real-transcript labels. A real-transcript-only training ablation would be needed to isolate how much of the reported F1 is attributable to the calibrated labels specifically.

Coordinator dependency. Revision quality depends on the coordinator model. A weaker coordinator may misdiagnose disagreement or introduce new ambiguities. Retry logic mitigates transient failures but not systematic ones. Because the coordinator is a frontier API model and is not bit-reproducible across runs, we expect the exact per-domain ceiling to vary by roughly one to two points between runs, based on typical run-to-run variance for this kind of model; we have not verified this band empirically, since, as noted immediately below, only a single calibration

run was performed. We accordingly do *not* claim the qualitative findings (the capability ceiling, degenerate-judge detection, and the three agreement regimes) are empirically robust to run-to-run variation—only that they are unlikely to be artifacts of a single anomalous coordinator call, given that each rests on a consistent diagnosis repeated across multiple iterations within the one run we did perform.

Single calibration run. The reported convergence trajectory (Table 2) reflects a single run of a non-deterministic coordinator, and no repeated-run robustness check was performed. In particular, Career & Self-Improvement’s iteration-2 gain (68.4% $\rightarrow$ 70.0%, 1.6 points) falls within the estimated ± 1 –2 point run-to-run variance band described above; that specific incremental result should therefore be read cautiously as a possible artifact of coordinator noise rather than confirmed as a genuine improvement from the rubric revision. Repeated runs would be needed to establish it as signal rather than noise, and more generally to establish empirically (rather than argue plausibly) that this paper’s qualitative findings are robust to coordinator variation.

No baseline comparison, partially addressed. We did not benchmark CRUCIBLE against manual rubric revision or simple self-refinement; these remain future work. A single frontier judge *is* now compared, both without a rubric and with the calibrated rubric (Section 5.8), so that specific gap is closed. Isolating the coordinator-mediated loop’s contribution from these still-untested baselines remains open.

Coordinator contribution: sensitivity-tested, not isolated. Section 5.8.4 now compares the original coordinator’s revision against two alternatives on Technology & AI—one given the same disagreement evidence, one given only the rubric—and finds the original coordinator outperformed both, while the disagreement-informed alternative underperformed the blind one. This is informative but does not isolate the coordinator’s causal contribution: both alternatives were generated by a different model (GPT-5.6 Sol, substituting for the unavailable Claude Sonnet coordinator), so the comparison cannot cleanly separate “the coordinator’s diagnostic reasoning is better” from “Claude Sonnet writes better surgical rubric edits than GPT-5.6 Sol for this task,” and only one alternative per condition was tried on one domain. A same-model ablation comparing coordinator-proposed revisions against alternative revisions of similar scope—ideally using Claude Sonnet for all conditions, or adding human-written alternatives—would more cleanly isolate whether the coordinator’s specific diagnosis matters. This remains future work.

Human-validation sample and rater depth. Our human validation (Section 5.6) now covers the full 60-transcript corpus, 20 per domain, which closes the sample-size limitation the earlier version of this work scoped as future work. The binding constraint is no longer the number of items but the number of raters: three per cohort, six per domain across the two rounds. The bootstrap intervals in Table 4 remain wide at that depth—General Education versus Technology & AI does not clear zero on the human side—so we present the reliability estimates as a ranking with stated uncertainty rather than as resolved effect sizes. Channel identity is recorded for this corpus; the channel-resampled intervals in Appendix G were computed on the round-1 30-item sample and were not recomputed at the larger size, and most channels contribute only one transcript to that sample—but the underlying 60-transcript corpus does have repeated channels (up to 5 from one channel), so this should not be read as ruling out channel-level dependence in general. Relatedly, for Career & Self-Improvement specifically (human AC1=0.33), the 2-of-3 majority vote used as “human consensus” throughout Sections 5.6, 5.7, and 5.8 is itself a noisy estimate given how weakly the three annotators agree with each other on this domain; annotator-level uncertainty is not propagated into the human-vs-CRUCIBLE comparison (Table 4) or the frontier-baseline comparison (Table 6), both of which treat the majority label as a fixed point estimate. Separately,

our human task uses an unaided-judgment design only (Section 5.6’s Design subsection): annotators were never shown the calibrated rubric. This tests whether CRUCIBLE’s labels align with unaided human intuition, not whether humans agree with the rubric’s specific operational criteria when shown it; a two-condition design (unaided plus rubric-guided) would additionally test construct validity of the rubric’s criteria themselves and is scoped as future work.

Held-out generalization is narrower than it may sound. Section 5.3’s positive result generalizes the rubric revision’s effect on *inter-judge agreement*, not on agreement with a human reference (no human labels exist for that 90-transcript corpus), and the “unseen-to-coordinator” claim rests on documented project workflow, not a logged, timestamped audit trail ruling out the corpus’s exposure during development. The two corpora also share 6 channels (Section 4), so this is document-held-out, not source-held-out. Most materially, the effect is driven disproportionately by one judge (Qwen 2.5 7B) shifting its label distribution toward the other two, largely static, judges—not by uniform three-way convergence toward more nuanced shared judgment—and the underlying majority-vote annotation changes on **0 of the 90** held-out transcripts between the v0 and final rubric (Section 5.3), so the generalization result should not be read as evidence the calibrated rubric produces more broadly reliable judgment on unseen data, or that it changes any actual output label on this set, only that it reliably shifts one judge’s individual predictions in a direction that happens not to move the already-fixed majority.

Frontier baseline sampling. GPT-5.6 Sol does not expose temperature control; our stability check (Section 5.8) characterizes label stability under the model’s default sampling rather than variance at a fixed, controlled temperature, and cannot separate genuine sampling noise from serving-side non-determinism.

Reasoning-before-label scope. The reasoning-before-label ablation (Section 5.8.3) uses a single, untuned prompt design tested across all three domains but not across alternative formulations; whether more carefully engineered reasoning prompts alter the pattern of raised inter-judge agreement without raised agreement with the human reference is untested.

Shared evaluation reference. The human validation, frontier-vs-human comparison, rubric-portability, reasoning-before-label, and the human-alignment portion of the judge-replacement analysis (Sections 5.6, 5.7, 5.8, 5.8.3, and 6.3) are all evaluated against the same 30-transcript human-validation set (or, for judge replacement, its Career subset). They test distinct manipulations but are related diagnostic analyses on one reference set, not independent replications in the statistical sense: a property specific to this particular reference set could in principle affect several of them jointly. The revision-sensitivity analysis (Section 5.8.4) and held-out evaluation (Section 5.3) use different corpora and are not subject to this specific concern, though each has its own scope limits noted above.

Reproducibility of the original run’s exact configuration. Before treating anything as unrecoverable, we checked available sources for the primary calibration run’s exact configuration. Recovered: the run’s timestamp (2026-05-31, from `calibration_summary.json` and corroborated by output file modification times); current digests, quantization (Q4_K_M for all three original judges), and context window (32,768 tokens for Qwen 2.5 7B and Mistral 7B, 131,072 for Llama 3.1 8B) via `ollama show`, which local file modification timestamps (“7 weeks” prior to this writing) are consistent with, but do not prove, matching what was used in the original run; and confirmation from the judge-calling code itself (`judge_agent.py`) that temperature was explicitly set to 0.0 and no context-window override was applied, so each model’s default context window applied throughout—ample for this task’s prompts (transcripts capped at 3,000 characters plus a short rubric, well under any

of these limits), which rules out context-window truncation as an explanation for Qwen’s specific misreadings, addressing one previously open question. Not recoverable: shell history containing the original `ollama pull` commands and their exact timestamps (none found); API billing or usage logs for the Claude Sonnet coordinator calls (not accessible in this environment); and, most importantly, a saved digest snapshot from *the original run itself* to diff against today’s digests, since none was captured at the time. Follow-up experiments in this paper (Sections 5.3, 6.3, and 5.8.4) were conducted under this closest recoverable configuration and should be interpreted as diagnostic extensions rather than exact controlled replications of the original run.

Runtime. At 17–20 minutes per iteration, a 10-iteration run requires nearly three hours. Parallelizing judge evaluation across models would reduce iteration time by roughly two-thirds.

8 Ethics Statement

This work’s human validation component (Section 5.6) involved data collection from human participants via Prolific. No formal institutional ethics review or exemption was sought; participation was governed solely by Prolific’s standard platform consent and terms. Participants were compensated at \$12.69/hour, above the platform minimum, for tasks with a 140-minute estimated completion window. No personally identifiable information beyond pseudonymous Prolific IDs was collected. Task descriptions accurately represented the work with no deceptive framing. All other experiments in this paper (rubric calibration, frontier baseline comparisons, the reasoning-before-label ablation, held-out evaluation, judge replacement, and revision-sensitivity analysis) involve no human subjects and use publicly available or synthetically generated transcript data.

Use of generative AI. Generative AI tools were used in this work for code generation and debugging of the experimental pipeline, and for writing refinement and editorial revision of the manuscript. All experimental design decisions, result interpretation, and scientific claims are the author’s own; all reported numbers derive from the released code and data, and the author reviewed and takes full responsibility for the final content.

9 Conclusion

CRUCIBLE demonstrates that annotation rubric calibration can be automated, after an initial rubric draft, through a coordinator-mediated iterative loop driven by inter-judge agreement signal, with no human reference labels used during optimization and no human in the loop after initialization (every retention, rollback, irreducibility-flagging, and stopping decision in the loop is programmatic; see Section 3.4). The system was built to solve a concrete problem—an earlier, differently-scoped pilot showed inconsistent, largely-uncalibrated judge disagreement (Introduction, Appendix B)—and raw weighted agreement using the local calibration panel reached 81.7%–96.7% on two domains, though chance-corrected agreement shows the genuine gain concentrates in Technology & AI. This paper’s central finding, however, concerns the criterion that decides when the loop has succeeded. Because the weighted agreement statistic the loop optimizes has a floor of 0.667, a threshold of $\tau = 0.80$ is satisfied at 40% unanimity, and the domain that crossed it did so at a chance-corrected agreement of 0.286 (Section 5.1.1). A human validation study spanning the full corpus, at double the sample size of our earlier report, does not support the human-alignment dissociation we previously described as preliminary: judge and human reliability rank the three domains identically, and match against human consensus fails to beat majority-class labelling in five of six cells (Section 5.6). Several interventions raise the optimized metric without moving agreement with the human reference—most visibly judge replacement, which raised inter-judge agreement from 0.700 to

0.883 while leaving human agreement bit-for-bit unchanged. As Section 6.3 details, that specific result is narrower than it first appears: the two retained judges already agreed with each other on 90% of the domain’s transcripts, so the panel’s majority-vote output was mechanically unchangeable by any third judge, and the null result on human alignment is consistent with, rather than independent proof of, the broader claim. The framework’s durable value is its diagnosis of disagreement structure, a check that this diagnosis’s effect on the inter-judge *agreement metric* generalizes past the calibration corpus even though the underlying majority-vote annotations do not change on the held-out set (Section 5.3), and the downstream classifier case study ($F1 = 0.871$) the calibrated rubrics enabled.

Beyond the specific numbers, CRUCIBLE surfaces one finding with broader implications: the capability ceiling, and its limits. Stagnation under coordinator-mediated revision is a diagnostic—it identifies that the limiting factor is plausibly related to a judge’s model-level capability rather than rubric underspecification, though our judge-replacement test cannot cleanly separate a genuine capability limitation from a structural property of a panel where two judges already agree most of the time, and motivates judge replacement or ensemble augmentation as directions worth testing over further revision. Section 6.3 tests judge replacement directly and shows it resolves the agreement symptom completely without changing the panel’s output at all, on the one domain and one replacement model tested, and by construction could not have changed the output on this specific panel—a result that sharpens rather than closes the diagnostic question. On a practical note, the columnar evaluation order (batching all transcripts per model before switching) reduced per-iteration model-load events from 180 to 3, making a full 10-iteration run tractable on an Apple Silicon MacBook with no discrete GPU (exact CPU/GPU offload configuration for the Ollama-served models was not recorded).

Code, calibration logs, versioned rubrics, and calibrated labels are publicly available at <https://github.com/biditdas18/rubric-calibration-agent>.

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A Rubric Calibration: Before and After

We show the initial (v0) and final calibrated rubric text for each domain. Only HIGH-signal criteria are shown for brevity; full rubrics (including LOW-signal criteria and conflict rules) are available in the released `calibrated_rubrics.json`.

General Education (v0 → v0, unchanged). General Education converged on iteration 1 with zero rubric changes; the initial rubric required no revision.

Technology & AI (v0 → v1). *v0*: “Names specific technologies, algorithms, libraries, or architectural patterns with accurate technical detail (e.g. ‘use Redis for session caching’, ‘transformer attention is $O(n^2)$ in sequence length’)”. *v1 (retained)*: “Names specific technologies, algorithms, data structures, or architectural patterns with accurate technical detail (e.g. stack-based $O(n)$ solution, Project Mainline module updates, M5 neural engine)”. The revision broadened “libraries” to “data structures” and replaced software-engineering-specific examples with domain-neutral technical examples, addressing Qwen 2.5 7B’s overly narrow reading of the original criterion; this single change raised agreement from 76.7% to 81.7%.

Career & Self-Improvement (v0 → v1 retained, v2 discarded). *v0*: “Contains at least 2 specific procedural steps with named methods, tools, or timelines (e.g. ‘File Form I-539 within 60 days’, ‘Use the STAR method for interviews’)”. *v1 (retained as final)*: “Contains at least 2 specific procedural steps with named methods, tools, timelines, or defined techniques—conceptual definitions alone do not qualify but named practices or described processes do”. This revision added an explicit exclusion clause to address Qwen 2.5 7B crediting abstract restatement as procedural content, raising weighted agreement from 68.4% to 70.0%—the domain’s best-observed result. A second revision (v2, not reproduced here as it was not retained) further tightened the co-occurrence condition between motivational language and procedural content; this regressed agreement to 68.4% and was discarded in favor of v1. The revisions shown above increase specificity rather than simplify toward keyword-matching, addressing the concern that rubric calibration might optimize for consensus by trivializing criteria rather than sharpening them.

B 3-Category Pilot Baseline: Full Figures

During development, an earlier, differently-scoped pilot annotation pipeline used a domain-agnostic rubric applied across a frontier chat-model ensemble (GPT-4, Gemini, Meta AI), reported in the companion SNR paper [24], Section 4.1. That pilot used a 3-category label space (HIGH/MID/LOW) later collapsed to binary via majority vote for downstream use, computed from the raw per-judge labels: 41.1% average pairwise agreement across the three judge pairs, and 16.7% full three-way unanimity (10/60 transcripts fully unanimous; 44/60 with exactly one matching pair; 6/60 with all three judges giving different labels). The 6/60 fully-split rate is possible because the label space has 3 categories, not 2 as in CRUCIBLE’s own metric—we do not attempt to reconcile the 41.1% and 16.7% figures via a binary-case formula, since that formula does not apply to a 3-category space; both figures are independently correct as computed directly from the raw per-judge labels.

Non-comparability. This pilot (3-category label space, a different model ensemble, pair-

wise/unanimity metrics) and CRUCIBLE’s own calibration metric (binary label space, the local judge panel, the weighted- A metric of Eq. 1) are measured in different label spaces with different models and are **not** directly comparable in magnitude. We present the pilot as qualitative motivating context—an independently-run experiment showing low, inconsistent agreement in an uncalibrated, differently-scoped setup—not as a quantitative pre/post baseline for CRUCIBLE’s own reported convergence numbers (76.7%–96.7% weighted A). No percentage-point improvement between the pilot and CRUCIBLE’s results is computed or implied anywhere in this paper.

C Columnar Evaluation Order: Full Derivation

The naive *row-wise* approach:

$$T_1 \rightarrow [\text{Llama, Mistral, Qwen}] \rightarrow T_2 \rightarrow [\text{Llama, Mistral, Qwen}] \rightarrow \dots$$

requires Ollama to load, run, and unload three model weight sets for every transcript. On consumer hardware with limited VRAM, each model switch incurs a cold-load penalty. Across 60 transcripts and 3 models per transcript, this produces 180 model-load events per iteration. In practice, this scheme could not complete a single iteration within 30 minutes and was cancelled.

The fix draws on a principle from columnar database design: group operations by column (model) rather than by row (transcript):

$$\underbrace{\text{Llama} \rightarrow 60 \text{ transcripts}}_{\text{col. 1}} \rightarrow \underbrace{\text{Mistral} \rightarrow 60 \text{ transcripts}}_{\text{col. 2}} \rightarrow \underbrace{\text{Qwen} \rightarrow 60 \text{ transcripts}}_{\text{col. 3}}$$

This reduces model-load events from 180 to 3 total per iteration. Each model loads once and remains resident in memory for its full 60-transcript pass. Result: each iteration completed in 17–20 minutes on a consumer MacBook (Apple Silicon, no discrete GPU; exact CPU/GPU offload configuration not recorded), making a 10-iteration calibration run (170.7 minutes total; the run terminated at the maximum-iteration cap of 10 iterations) fully tractable.

D Held-Out Rubric Evaluation: Full Detail

Full per-domain metrics and per-judge label prevalence (HIGH rate) underlying Table 3 (Section 5.3):

Domain	Cond.	All-agree	Pairwise	κ	AC1	Prevalence (L/M/Q)
Career & SI	v0	16.7%	44.4%	−0.385	+0.072	100% / 100% / 17%
Career & SI	Final	36.7%	57.8%	−0.268	+0.367	100% / 100% / 37%
Tech & AI	v0	53.3%	68.9%	−0.184	+0.578	100% / 100% / 53%
Tech & AI	Final	63.3%	75.6%	−0.139	+0.689	100% / 100% / 63%
Gen. Education	Both	96.7%	97.8%	−0.011	+0.977	100% / 100% / 97%

L/M/Q = Llama 3.1 8B / Mistral 7B / Qwen 2.5 7B. Llama and Mistral predict HIGH on essentially every transcript in both revised domains under both rubric versions; Qwen is the only judge whose prevalence shifts, and by an amount that tracks the reported weighted- A improvement closely. Judge calls: 90 transcripts $\times$ 2 rubric versions $\times$ 3 judges = 540, single inference pass each, temperature = 0.0, cached and resumable; script and raw per-call cache are released with this paper’s code.

E Irreducible-Case Audit

Independent audit: infeasible. We attempted to independently re-examine the coordinator’s irreducible-case calls (Section 5 originally named two specific Technology & AI transcripts) against the actual per-case criterion quotes the coordinator saw. No such log survives: inspection of `calibration_loop.py` and `coordinator_agent.py` confirms only three files are ever persisted (`calibrated_rubrics.json`, `calibrated_labels.csv`, `calibration_summary.json`); flagged-irreducible transcript indices are tracked only as an in-memory Python set during the run and never written to disk. The specific transcript identities named in an earlier draft of this paper could not be verified against any surviving artifact and have been removed from the main text accordingly.

Sensitivity check: completed. Once a transcript is flagged irreducible, `calibration_loop.py` excludes it from that domain’s active evaluation pool for all subsequent iterations (confirmed by code inspection), so it stops contributing to the domain’s tracked weighted- A denominator from that point on (the “excluded” treatment, i.e., the number actually reported during the run). We compare this to an “included” treatment: the full re-relabeling of all 20 transcripts per domain under the final rubric (`calibrated_labels.csv`), which counts every transcript regardless of irreducible-flag status. This is the most direct recoverable proxy for “included,” not a reconstruction of the original two flagged cases, since their specific identities are not recoverable (above).

Domain	Irreducible count	Excluded (reported)	Included (proxy)
General Education	0	96.7%	96.7%
Technology & AI	2	81.7%	81.7%
Career & Self-Impr.	0	68.4%	70.0% [†]

[†]Career’s difference is *not* an irreducible-case effect (its count is 0, so this check is a structural no-op there); it reflects the pre-existing v1-retained-vs-v2-discarded discrepancy already discussed in Section 5 and Table 2.

For Technology & AI, the domain with irreducible cases, the convergence conclusion (converged, 81.7%, above the 0.80 threshold) is identical under both treatments. As an upper-bound reference, 11 of the 20 transcripts still show inter-judge disagreement in the full final-rubric relabel (global indices 24, 25, 28, 30, 31, 33, 35, 36, 37, 38, 39)—a superset of whichever 2 were originally flagged irreducible, not those cases specifically.

F Prolific Rejection Sensitivity Check

One of nine human-validation submissions (Technology & AI domain) was rejected and replaced. Cross-referencing the pre- and post-replacement raw batch exports against the officially recorded consensus (`human_validation_ground_truth.csv`) by testing which of the four possible 3-of-4 annotator-response subsets exactly matches the recorded consensus multiset on all 10 non-attention-check items identifies the rejected annotator unambiguously (10/10 exact match for one subset; 6–7/10 for every other), independently confirmed against the Prolific dashboard’s Rejected tab (single rejected participant, matching the rejected submission’s internal record; completion time 39:06 against this batch’s 140:00 median).

Recomputing this domain’s human consensus with the rejected annotator’s original responses reinstated in place of the replacement annotator’s changes **zero** of the 10 item-level consensus labels. Local-panel-vs-human agreement for Technology & AI is identical under both the official and re-

instated consensus (agreement 50.0%, Cohen’s $\kappa = -0.19$, Gwet’s AC1 = 0.20, both conditions). The domain-level conclusions reported in Section 5.6 do not depend on this exclusion.

G Channel-Clustered Confidence Intervals

Full detail underlying the channel-clustering comparison summarized in Limitations. Channel identity for the 60-transcript corpus is available via `video_selection_60.csv` (SNR-Detector repository), joined to this corpus by verified positional alignment. Bootstrap intervals below use 10,000 resamples; “channel-resampled” resamples at the channel level (with all of a sampled channel’s transcripts included together) rather than the transcript level.

Domain	# channels	Point	Transcript-CI	Channel-CI
Career & Self-Impr.	10	50.0%	[20.0%, 80.0%]	[20.0%, 80.0%]
General Education	9	90.0%	[70.0%, 100.0%]	[66.7%, 100.0%]
Technology & AI	8	50.0%	[20.0%, 80.0%]	[22.2%, 77.8%]
Overall (27 unique channels of 30)		63.3%	[46.7%, 80.0%]	[46.7%, 79.3%]

Career & Self-Improvement’s 10 sampled transcripts happen to come from 10 distinct channels, so its channel- and transcript-resampled intervals are identical by construction. General Education and Technology & AI each have one repeated channel within their 10-item sample, producing a small widening or shift at one bound (General Education’s lower bound moves from 70.0% to 66.7%; Technology & AI’s interval shifts from [20.0%, 80.0%] to [22.2%, 77.8%]). The overall effect across all 30 items is small (upper bound 80.0% $\rightarrow$ 79.3%). This should be read as evidence channel clustering has a small effect *on this particular 30-item sample*, not as evidence it would be negligible at different sample sizes or samples with heavier channel repetition, such as the full 60-transcript corpus (up to 5 transcripts from one channel) or the 90-transcript training-supplement corpus (up to 7 from one channel).